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Key Points:

- Using typical magnetospheric plasma parameters, a series of simulations is performed to quantify the repetition period of chorus elements
- The repetition period is controlled by energetic-electron injection and nonlinear wave growth, with the latter slightly more influential
- Regression analysis shows that repetition period increases with electron injection timescale but decreases with nonlinear growth rate

Supporting Information:

Supporting Information may be found in the online version of this article.

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Chorus Element Repetition: Roles of Energetic Electron Injection and Nonlinear Wave Growth

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Abstract Chorus waves are coherent whistler-mode emissions in Earth's magnetosphere and play a key role in the acceleration and loss of radiation belt electrons. Their spectra commonly consist of discrete and repetitive elements, yet the mechanism governing this repetition is not fully understood. Using typical magnetospheric plasma parameters, we perform simulations to investigate the excitation of repetitive chorus elements. The repetition period is quantified and its dependence on plasma parameters and the background magnetic-field configuration is examined. The results identify two factors controlling chorus element repetition: external energetic-electron injection and intrinsic nonlinear wave growth. Energetic-electron injection replenishes free energy, whereas nonlinear growth converts this free energy into wave energy. Specifically, the repetition period increases with the electron injection timescale but decreases with the nonlinear growth rate. Standardized regression coefficients indicate a slightly stronger influence of nonlinear wave growth on the repetition period.

Plain Language Summary Chorus waves are electromagnetic emissions commonly observed in Earth's magnetosphere. Their spectra typically consist of discrete and repetitive elements, but the physical mechanism underlying this repetition remains incompletely understood. In this study, we perform simulations using plasma parameters representative of Earth's magnetosphere to investigate the excitation of repetitive chorus elements and quantify the dependence of repetition period on key plasma parameters. The results show that chorus element repetition is controlled by two factors: energetic-electron injection and nonlinear wave growth, which play complementary roles in replenishing and releasing free energy in the energetic-electron velocity distribution, respectively. Regression analysis further shows that the repetition period increases with the electron injection timescale but decreases with the nonlinear growth rate, with nonlinear wave growth playing the dominant role. Because repetitive chorus elements have also been observed in other planetary and lunar plasma environments, the mechanisms identified here may have broader relevance beyond Earth.

1. Introduction

Quasi-periodic wave repetition has been observed in a wide range of plasma emissions throughout Earth's magnetosphere, including chorus waves (Agapitov et al., 2017; Demekhov et al., 2020; Gao et al., 2022; Li, Bortnik, et al., 2011; Li, Thorne, et al., 2011; Santolík et al., 2003; Xia et al., 2020), electron cyclotron harmonic (ECH) waves (Shen et al., 2021), electromagnetic ion cyclotron (EMIC) waves (Liu et al., 2019), and magnetosonic waves (Boardsen et al., 2026; Sun et al., 2020). Among these emissions, chorus waves have attracted particular attention because of their crucial role in regulating electron dynamics within the hazardous radiation belt (Agapitov et al., 2014; Goyal et al., 2018; Hosseini et al., 2019; Lejosne et al., 2022; Meredith et al., 2003; Summers et al., 1998; Thorne et al., 2013).

Chorus waves are coherent whistler-mode emissions characterized by discrete and repetitive frequency chirps (Burtis & Helliwell, 1969; Harid et al., 2014; Santolík et al., 2003; Tsurutani & Smith, 1974), commonly referred to as repetitive chorus elements. Such repetition has been reported in both rising-tone (Gao et al., 2022; He et al., 2023; Shue et al., 2015, 2019) and falling-tone chorus (X. Chen et al., 2026). The repetition of chorus elements plays an important role in modulating pulsating aurora in the upper ionosphere (H. Chen et al., 2025; R. Chen et al., 2024; Hosokawa et al., 2020; Miyoshi et al., 2015; Ozaki et al., 2018). Although chorus element excitation is widely attributed to nonlinear wave-particle interactions (H. Chen et al., 2023, 2024; R. Chen et al., 2025; Demekhov et al., 2017; Gołkowski et al., 2019; Nunn et al., 1974; Omura et al., 1991; Tao et al., 2021; Vomvoridis et al., 1982; Wang et al., 2024; Zonca et al., 2022), the physical mechanism governing this repetition is not fully understood.

Recent theoretical studies have proposed that chorus element repetition is associated with the injection of energetic electrons during geomagnetically active periods (H. Chen et al., 2025; Lu et al., 2021). These electrons drift through the magnetosphere and continuously replenish the free energy available for wave growth, enabling the generation of repetitive chorus elements. This mechanism is supported by satellite observations showing that the chorus repetition period, defined as the time interval between adjacent elements, is inversely correlated with energetic-electron drift velocity (Gao et al., 2022). It is further supported by laboratory experiments using the Large Plasma Device (LAPD), in which controlled electron injection generates repetitive whistler-mode emissions (D. Ma et al., 2025).

However, observations also reveal a pronounced day-night asymmetry in the chorus repetition period, which is typically longer on the dayside (Gao et al., 2022; Shue et al., 2015), indicating that energetic-electron injection alone cannot fully account for the observed repetition. Statistical analyses further show that the chorus repetition period depends not only on energetic-electron distribution but also on background plasma conditions, including magnetic-field strength and plasma density (Shue et al., 2015). Nevertheless, because these parameters often vary simultaneously, their individual effects on repetition cannot be isolated from observations.

In this study, we perform simulations to investigate the generation of repetitive rising-tone chorus elements under typical plasma conditions in Earth's magnetosphere. We first quantify the repetition period in a nominal case and then conduct a parameter survey to examine its dependence on plasma parameters and the background magnetic-field configuration. The results identify two factors controlling chorus element repetition: external energetic-electron injection and intrinsic nonlinear wave growth. Further analysis indicates a slightly stronger influence of nonlinear wave growth on the repetition period.

2. Simulation Model and Initial Setup

The simulations are conducted in the meridian plane using the general curvilinear plasma simulation code (GCPIC) model (Lu et al., 2019; Wang et al., 2024), which solves Maxwell's equations and updates particle dynamics in dipole coordinates. The one-dimensional simulations along the magnetic field line are performed. The background magnetic field $\mathbf{B}_0$ is prescribed as a dipole that can be compressed or stretched. The initial magnetic field components in Cartesian coordinates are given by

$$B_{0x} = -B_E R_E^3 \frac{3xz}{r^5}, \quad (1)$$

$$B_{0y} = -B_E R_E^3 \frac{3yz}{r^5}, \quad (2)$$

$$B_{0z} = -B_E R_E^3 \frac{3z^2 - r^2}{r^5}, \quad (3)$$

with

$$r = \sqrt{(\eta x)^2 + y^2 + z^2}. \quad (4)$$

Here $B_E = 31,200$ nT denotes the equatorial surface magnetic field strength, and $R_E = 6,371$ km is the Earth's radius. The parameter η controls the dipole geometry, where $\eta = 1$ corresponds to a normal dipole, $\eta < 1$ to a stretched configuration on the nightside, and $\eta > 1$ to a compressed configuration on the dayside. A schematic illustration of how η modifies the dipole configuration is provided in Figure 1 of Kong, Gao, Ke, et al. (2023).

The system includes three particle populations: energetic electrons, cold electrons, and protons. Both electron populations are treated kinetically using the relativistic Lorentz force, while protons are assumed to be immobile. At the magnetic equator, energetic electrons are initialized with a loss-cone distribution constructed by subtracting two bi-Maxwellian components (Zenitani & Nakano, 2023):

$$f(u_{\parallel}, u_{\perp}) = \frac{n_{heq}}{(2\pi)^{3/2} U_{\parallel} U_{\perp}^2 (1 - \rho\kappa)} \exp\left(-\frac{u_{\parallel}^2}{2U_{\parallel}^2}\right) \cdot \left[\exp\left(-\frac{u_{\perp}^2}{2U_{\perp}^2}\right) - \rho \exp\left(-\frac{u_{\perp}^2}{2\kappa U_{\perp}^2}\right) \right], \quad (5)$$

where $\rho = 1$ and $\kappa = 0.3$ are constants, n_{heq} is the equatorial energetic-electron density, $U_{\parallel} = \gamma v_{\parallel}$, $U_{\perp} = \gamma v_{\perp}$, $u_{\parallel} = \gamma v_{\parallel}$, and $u_{\perp} = \gamma v_{\perp}$. Here, $v_{\parallel}$ and $v_{\perp}$ are the parallel and perpendicular thermal velocities, $v_{\parallel}$ and $v_{\perp}$ are the corresponding velocity components, and γ is the Lorentz factor. The electron distribution along the field line is obtained from Liouville's theorem (Summers et al., 2012). Cold electrons follow a Maxwellian distribution.

The injection of energetic electrons is modeled by modifying the Vlasov equation with a source term:

$$\frac{\partial f}{\partial t} + \dot{q} \frac{\partial f}{\partial q} + \dot{\mathbf{u}} \frac{\partial f}{\partial \mathbf{u}} = -\frac{f - f_0}{\tau_D}, \quad (6)$$

where f and f_0 are the instantaneous and initial distribution functions, respectively, q is the curvilinear coordinate along the field line, and overdots denote time derivatives. Equation 6 describes continuously injection rather than injection at discrete time intervals. The parameter τ_D defines the characteristic injection timescale over which the energetic-electron distribution is replenished toward f_0 . In our simulations, τ_D is assumed independently of electron energy. Further details are given in Lu et al. (2021).

The simulations adopt typical plasma parameters at $L \sim 6$. In the nominal case, the equatorial background magnetic-field strength is $B_{e0} = 144.4$ nT ($=B_E/L^3$), and the plasma density is $n_e = 5$ cm $^{-3}$ (Denton et al., 2004; Fu et al., 2014). These values give a ratio of $\omega_{pe}/\Omega_{e0} = 5$, where $\omega_{pe} = \sqrt{n_e e^2 / (m_e \epsilon_0)}$ is the plasma frequency and $\Omega_{e0} = e B_{e0} / m_e$ is the equatorial electron gyrofrequency. Here, e , m_e , and ϵ_0 denote the elementary charge, electron mass, and vacuum permittivity. At the magnetic equator, the number density of energetic electrons is $n_{heq} = 0.09$ cm $^{-3}$ (Rasinskaite et al., 2025). The parallel temperature is $m_e U_{\parallel}^2 = 20$ keV (Li, Bortnik, et al., 2011; Li, Thorne, et al., 2011; Rasinskaite et al., 2025), and the temperature anisotropy is $U_{\perp}^2 / U_{\parallel}^2 = 1.5$ (Fu et al., 2014). The electron injection timescale is set to $\tau_D = 5,000 \Omega_{e0}^{-1}$. The temperature of cold electrons is 1 eV. A normal dipole configuration is used with $\eta = 1$. Additional cases are constructed by varying one parameter at a time from the nominal case. The specific parameters are described in Section 3.2.

The simulation domain is constructed at $L = 6$ and spans magnetic latitudes of $\sim \pm 30^\circ$, with reflecting boundary conditions applied to particles and absorbing boundary conditions imposed on fields. It is discretized with 5,000 grids along the field line, with a grid spacing of $0.26 V_{Ae} / \Omega_{e0}$, where $V_{Ae} = B_{e0} / \sqrt{\mu_0 n_e m_e}$ is the electron Alfvén speed and μ_0 is the vacuum permeability. Each grid contains 800 particles per species. The time step is $0.03 \Omega_{e0}^{-1}$. The low-noise δf algorithm is employed.

3. Simulation Results

3.1. Nominal Case Analysis

We first present the repetitive chorus elements in the nominal case and quantify the repetition period T_{rep} . Figure 1a shows the waveform of δB_{φ} at a magnetic latitude of $\lambda = 5^\circ$, and Figure 1b shows the corresponding wave spectrum of $\delta B = \sqrt{\delta B_r^2 + \delta B_{\varphi}^2}$. Here, δB_r and δB_{φ} are the two components perpendicular to the background magnetic field in the local field-aligned dipole coordinate system. The r direction lies in the meridional plane, whereas the φ direction is azimuthal. This latitude is chosen because in situ chorus observations are commonly made near the magnetic equator (Li et al., 2009; J. Ma et al., 2022). At this latitude, the background magnetic-field gradient is $\partial B_0 / \partial h = 5.97 \times 10^{-4} B_{e0} \Omega_{e0} / V_{Ae0}$, where B_0 is the local background magnetic-field strength and h is the distance along the field line to the magnetic equator. Chorus waves are self-consistently excited by energetic electrons and contain three rising-tone elements with frequencies increasing from ~ 0.2 to $\sim 0.7 \Omega_{e0}$. To quantify T_{rep} , an amplitude threshold of $0.004 B_{e0}$ is adopted. For each element, the interval over which the wave amplitude exceeds this threshold is determined, with its start and end times marked and connected by a white dashed line. The separation ΔT between adjacent elements is defined as the difference between their start times, and the repetition period T_{rep} is calculated as the average ΔT over the entire simulation. For the nominal case, $T_{\text{rep}} = 5,895 \Omega_{e0}^{-1}$, corresponding to ~ 0.23 s, which falls within the typical observational range of ~ 0.1 – 0.9 s (Gao et al., 2022; Shue et al., 2015).

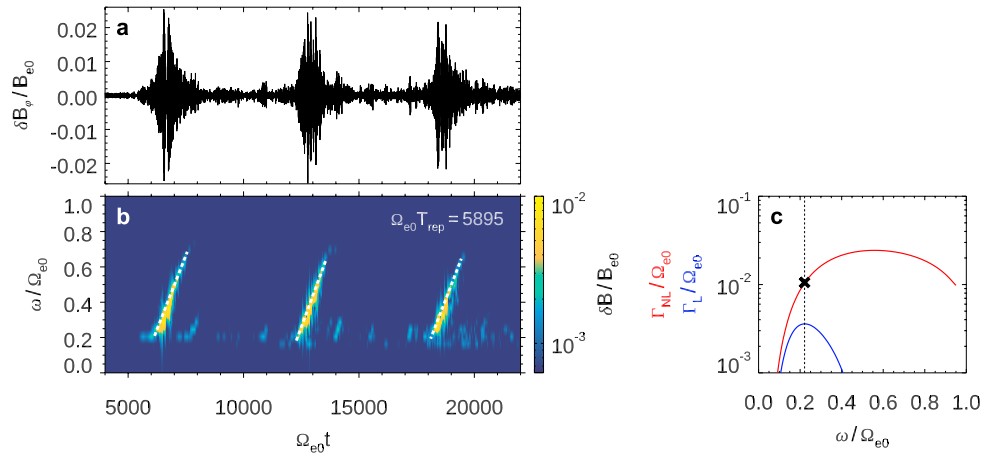


Figure 1. Nominal case showing repetitive chorus elements. (a) Waveform of $\delta B_\phi / B_{e0}$. (b) Wave spectrum of $\delta B / B_{e0}$, with each element denoted by a white dashed line and the corresponding repetitive period T_{rep} labeled. (c) Nonlinear growth rate $\Gamma_{NL} / \Omega_{e0}$ (red) and linear growth rate Γ_L / Ω_{e0} (blue). The dashed line marks the frequency at which Γ_L reaches its maximum, and the cross denotes the corresponding Γ_{NL} value.

Chorus elements are excited through nonlinear growth initiated from seed waves generated by linear instability, whose frequency corresponds to that of the maximum linear growth rate (Omura, 2021). Once the seed-wave amplitude exceeds the nonlinear threshold, coherent chorus elements are triggered through nonlinear wave-particle interactions, accompanied by frequency chirping from the seed-wave frequency (Vomvouridis et al., 1982; Wang et al., 2024; Zonca et al., 2022). To determine whether nonlinear processes also control chorus element repetition, we quantify the nonlinear growth rate and examine its relationship with T_{rep} .

The nonlinear growth rate Γ_{NL} for chorus waves is defined as (Omura, 2021)

$$\Gamma_{NL} = \frac{Q\omega_{ph}^2 v_g}{2\gamma U_{||}} \left(\frac{\xi}{\omega\Omega_w} \right)^{1/2} \left(\frac{\chi U_{\perp}}{\pi c} \right)^{3/2} \exp\left(-\frac{\gamma^2 v_c^2}{2U_{||}^2}\right) \quad (7)$$

where

$$\omega_{ph} = \sqrt{\frac{n_{heq} e^2}{m_e \epsilon_0}}, \quad (8)$$

$$\xi = \frac{\sqrt{\omega(\Omega_{e0} - \omega)}}{\omega_{pe}}, \quad (9)$$

$$\chi = \sqrt{1 - \frac{\omega^2}{c^2 k^2}}, \quad (10)$$

and

$$\Omega_w = \frac{ec}{0.4m_e \omega} \frac{s_2}{s_0} \frac{\partial B_0}{\partial h}. \quad (11)$$

Here,

$$s_0 = \frac{\xi c}{\chi v_{\perp}}, \quad (12)$$

$$s_2 = \frac{1}{2\xi\chi} \left\{ \frac{\gamma\omega}{\Omega_{e0}} \left(\frac{v_\perp}{c} \right)^2 - \left[2 + \frac{\omega}{\Omega_{e0}} \frac{\chi^2(\Omega_{e0} - \gamma\omega)}{\Omega_{e0} - \omega} \right] \frac{v_c v_p}{c^2} \right\}. \quad (13)$$

In addition, c is the speed of light, k is the wave number, v_g and v_p are the group and phase velocities, and the cyclotron resonant velocity is defined as $v_c = (\omega - \Omega_{e0}/\gamma)/k$. The constant Q is set to 0.5, and $U_\perp$ is taken to be $2.5 V_{Ae0}$.

Figure 1c shows Γ_{NL} as a function of wave frequency ω (red curve), calculated using the plasma parameters at the magnetic equator and the background magnetic-field gradient at $\lambda = 5^\circ$. The linear growth rate Γ_L , derived from the equatorial plasma parameters, is overplotted for reference (blue curve). To ensure a consistent comparison across different cases, the Γ_{NL} at the seed-wave frequency is recorded. This frequency corresponds to the wave frequency at the start time of each element, which is used to quantify the repetition period. For the nominal case, the seed-wave frequency is $0.22 \Omega_{e0}$ (indicated by the black dashed line), which agrees with the minimum frequency of the chorus elements at $\sim 0.2 \Omega_{e0}$. The corresponding nonlinear growth rate is $\Gamma_{NL} = 1.05 \times 10^{-2} \Omega_{e0}$, as marked by the black cross.

3.2. Parametric Study

The simulations can isolate the effects of individual plasma parameters. We therefore perform a series of parameter scans to investigate the dependence of T_{rep} on the injection timescale τ_D , energetic-electron density n_{heq} , plasma density n_e , and background magnetic-field gradient $\partial B_0/\partial h$. Specifically, τ_D is varied from 2,000 to 8,000 Ω_{e0}^{-1} in increments of 500 Ω_{e0}^{-1} ; n_{heq} from 0.08 to 0.11 cm^{-3} in increments of 0.0025 cm^{-3} ; and n_e from 4.5 to 7.5 cm^{-3} in increments of 0.25 cm^{-3} . The magnetic-field gradient $\partial B_0/\partial h$ is varied by changing η from 0.95 to 1.05 in increments of 0.01, making $\partial B_0/\partial h$ decreasing from 6.94×10^{-4} to $5.14 \times 10^{-4} B_{e0} \Omega_{e0}/V_{Ae0}$. Only one parameter is varied at a time, with all other parameters held at their nominal values. For each simulation, Γ_{NL} at the seed-wave frequency is quantified using the procedure described in Figure 1c.

The selected parameter ranges are constrained by observations and empirical models. A range of $\tau_D \sim 3,000$ – $15,000 \Omega_{e0}^{-1}$ has been shown to yield repetition periods consistent with observations (Gao et al., 2022). Observational statistics indicate that n_{heq}/n_e is typically on the order of 1% near $L = 6$ (Gao et al., 2014; Rasinskaite et al., 2025), and n_e generally ranges from ~ 1 to 10 cm^{-3} (Denton et al., 2004). The geomagnetic field model of Tsyganenko and Sitnov (2005) suggests that at $\eta \sim 1.1$ on the dayside under solar-wind compression and values as low as ~ 0.7 on the nightside due to field-line stretching at $L \sim 6$. Here, η is varied by only $\pm 5\%$ about its nominal value to minimize changes in the particle velocity distribution.

Figure 2 shows the spectrum of repetitive chorus elements at $\lambda = 5^\circ$ for (a1–a3) $\tau_D = 2,000, 4,000$, and $6,000 \Omega_{e0}^{-1}$; (b1–b3) $n_{\text{heq}} = 0.0825, 0.0925, 0.1025 \text{ cm}^{-3}$; (c1–c3) $n_e = 4.75, 5.25$, and 5.75 cm^{-3} ; and (d1–d3) $\partial B_0/\partial h = 6.52 \times 10^{-4}, 6.12 \times 10^{-4}$, and $5.75 \times 10^{-4} B_{e0} \Omega_{e0}/V_{Ae0}$, corresponding to $\eta = 0.97, 0.99$, and 1.01 , respectively. The repetition period T_{rep} in each case is quantified using the method illustrated in Figure 1b. Overall, T_{rep} increases with τ_D and $\partial B_0/\partial h$ but decreases with n_{heq} and n_e within the parameter ranges considered here.

We then quantify the dependence of T_{rep} on each of the four parameters separately. Figure 3a shows T_{rep} as a function of τ_D , with each blue dot representing an individual simulation case. T_{rep} increases nearly linearly with τ_D , consistent with previous observations that faster electron drift results in more rapid electron injection and thus a shorter repetition period (Gao et al., 2022). Linear regression gives $\Omega_{e0} T_{\text{rep}} = 0.99 \Omega_{e0} \tau_D + 636.75$, with a coefficient of determination $R^2 = 0.98$. Here, R^2 is defined as $R^2 = 1 - \text{SSE}/\text{SST}$, where SSE and SST are the sums of squared deviations from the fitted values and the mean, respectively. The R^2 value close to one indicates an excellent fit.

Variations in n_{heq} lead to corresponding changes in the nonlinear growth rate Γ_{NL} . We therefore examine the dependence of T_{rep} on the inverse nonlinear growth rate Γ_{NL}^{-1} . Based on Equations 7 and 8, $\Gamma_{NL} \propto n_{\text{heq}}$. Figure 3b show T_{rep} as a function of $(n_{\text{heq}}/n_{\text{heq}N})^{-1}$, where $n_{\text{heq}N} = 0.018 \text{ cm}^{-3}$ denotes the energetic-electron density in the nominal case. This normalization makes the horizontal-axis variable dimensionless. T_{rep} increases nearly linearly

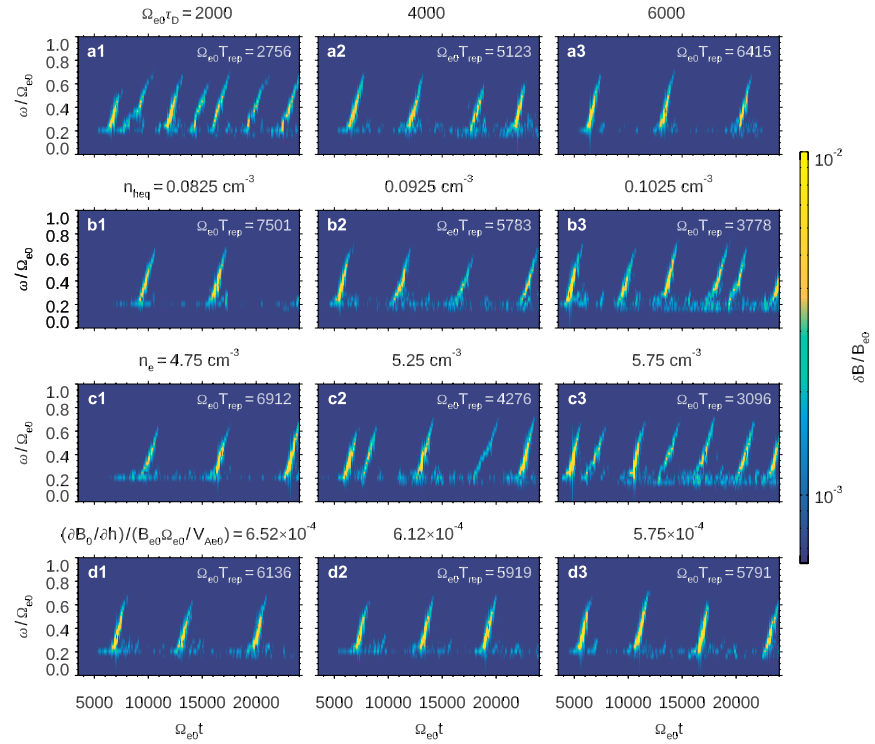


Figure 2. Parametric scan of chorus element repetition at $\lambda = 5^\circ$. Repetition period T_{rep} for varying (a1–a3) injection timescale τ_D , (b1–b3) energetic-electron density n_{heq} , (c1–c3) plasma density n_e , and (d1–d3) the background magnetic-field gradient $\partial B_0 / \partial h$. The corresponding T_{rep} value is indicated in each panel.

with $(n_{heq}/n_{heqN})^{-1}$, indicating an approximately proportional dependence on Γ_{NL}^{-1} . Linear regression yields $\Omega_{e0}T_{rep} = 1.86 \times 10^4 (n_{heq}/n_{heqN})^{-1} - 1.27 \times 10^4$ (black line), with $R^2 = 0.98$. This result shows that, in addition to electron injection timescale, the nonlinear growth rate also plays an important role in determining T_{rep} .

Γ_{NL} also depends on the plasma density n_e . This dependence is nontrivial because n_e affects both the normalized energetic-electron density n_{heq}/n_e and the wave dispersion relation. Using the simulation parameters, we numerically evaluate Γ_{NL} as a function of n_e (Figure S1 in Supporting Information S1) and obtain the empirical fit $\Gamma_{NL}(n_e)/\Omega_{e0} = 10^{-2} [1.14 - 1.70 \exp(-2.94 n_e/n_{eN})]$, where $n_{eN} = 5 \text{ cm}^{-3}$ is the nominal plasma density. Figure 3c shows that T_{rep} increases with Γ_{NL}^{-1} . The resulting fit is $\Omega_{e0}T_{rep} = 4.13 \times 10^2 [\Gamma_{NL}(n_e)/\Omega_{e0}]^{-1} - 4.20 \times 10^4$, with $R^2 = 0.97$.

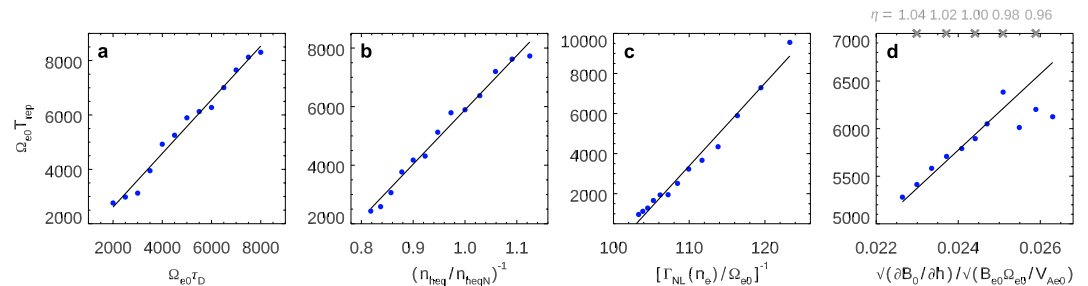


Figure 3. Quantitative dependence of T_{rep} on simulation parameters. Relationships between $\Omega_{e0}T_{rep}$ and (a) $\Omega_{e0}\tau_D$, (b) $(n_{heq}/n_{heqN})^{-1}$, (c) $[\Gamma_{NL}(n_e)/\Omega_{e0}]^{-1}$, (d) $(\partial B_0 / \partial h)^{1/2} / (B_{e0} \Omega_{e0} / V_{Ae0})^{1/2}$. Blue dots denote individual simulation cases, and black lines show linear fits. The horizontal-axis variables in panels (b–d) are proportional to Γ_{NL}^{-1} . Gray crosses in panel (d) denote the corresponding values of η .

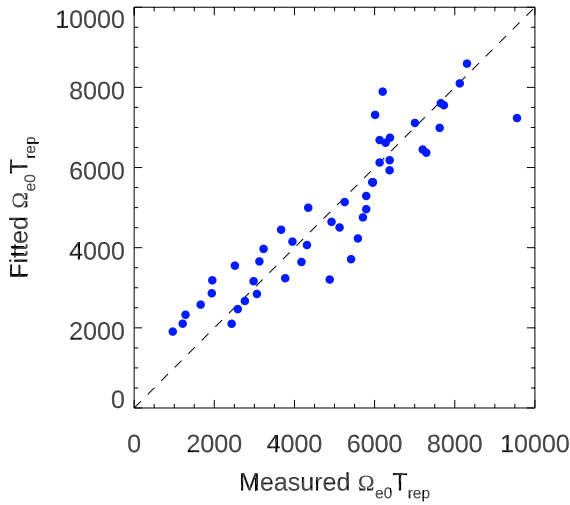


Figure 4. Comparison between the simulated and fitted repetition period. Blue dots represent all individual simulation cases, with the simulated $\Omega_{e0}T_{\text{rep}}$ on the horizontal axis and the values predicted by the empirical relation on the vertical axis. The dashed diagonal denotes one-to-one agreement.

$C_{\Gamma} = 2.78 \times 10^2$, and $C_0 = -3.44 \times 10^4$, with $R^2 = 0.82$. Figure 4 compare the measured T_{rep} with the fitted values, showing good overall agreement.

In this study, τ_D spans a broad range, whereas Γ_{NL}^{-1} varies over a relatively narrow range. This difference arises because Γ_{NL} cannot be controlled directly but is instead varied through plasma parameters, whose values are constrained by the conditions favorable for chorus-element excitation identified by X. Chen et al. (2026). Nevertheless, the variation in Γ_{NL} still produces a measurable change in T_{rep} (Figure 2b–2d). Therefore, within the parameter ranges examined here, the linear relationship described by Equation 14 is applicable.

The relative contributions of energetic-electron injection and nonlinear wave growth to repetition period are further assessed. Because τ_D and Γ_{NL}^{-1} differ in scale and range, the corresponding regression coefficients C_{τ} and C_{Γ} cannot be compared directly. Standardized regression coefficients are commonly used to compare the relative effects of different predictors (Mayer & Younger, 1974; van Ginkel, 2020) and are defined as

$$C_{\tau}^* = C_{\tau} \frac{\sigma_{\tau}}{\sigma_{T_{\text{rep}}}}, \quad (15)$$

$$C_{\Gamma}^* = C_{\Gamma} \frac{\sigma_{\Gamma}}{\sigma_{T_{\text{rep}}}}, \quad (16)$$

where σ_{τ} , σ_{Γ} , and $\sigma_{T_{\text{rep}}}$ are the standard deviations for τ_D , Γ_{NL}^{-1} , and the measured T_{rep} , respectively. The resulting values are $C_{\tau}^* = 0.46$ and $C_{\Gamma}^* = 0.78$. The two coefficients are comparable in magnitude, with the larger C_{Γ}^* indicating a slightly stronger influence of nonlinear wave growth on T_{rep} within the parameter ranges considered.

4. Discussions

The simulations conducted in this study show that variations in the background magnetic-field configuration modify the nonlinear growth rate and thereby affect the repetition period T_{rep} . In Figure 3d, however, T_{rep} exhibits a weak negative correlation with $(\partial B_0/\partial h)^{1/2}$ when $(\partial B_0/\partial h)^{1/2}$ exceeds $\sim 0.025 (B_{e0}\Omega_{e0}/V_{Ae0})^{1/2}$. This trend seems inconsistent with nonlinear theory, which gives $\Gamma_{NL} \propto (\partial B_0/\partial h)^{-1/2}$ and therefore predicts a longer T_{rep} for larger $(\partial B_0/\partial h)^{1/2}$. The discrepancy arises because varying the field-configuration parameter η changes not only the magnetic-field gradient but also the field-line curvature. Specifically, decreasing η increases the gradient and produces a more stretched field-line configuration characteristic of the nightside magnetosphere. Nevertheless, such curvature effects are not considered in existing nonlinear theories.

Based on Equations 7 and 11, the dependence of Γ_{NL} on magnetic-field gradient can be expressed as $\Gamma_{NL} \propto (\partial B_0/\partial h)^{-1/2}$, indicating that a larger magnetic-field gradient results in a lower nonlinear growth rate. Figure 3d shows that T_{rep} generally increases with $(\partial B_0/\partial h)^{1/2}$. A linear fit gives $\Omega_{e0}T_{\text{rep}} = 4.03 \times 10^4 (\partial B_0/\partial h)^{1/2} / (B_{e0}\Omega_{e0}/V_{Ae0})^{1/2} - 3.84 \times 10^4$, with $R^2 = 0.58$. The relatively weak correlation arises because the linear trend holds only at small gradients with $(\partial B_0/\partial h)^{1/2} < \sim 0.025 (B_{e0}\Omega_{e0}/V_{Ae0})^{1/2}$. For larger gradients, the simulated T_{rep} values are lower than those predicted by the linear fit. The underlying mechanism is discussed in Section 4.

Chorus element repetition is jointly controlled by external energetic-electron injection and intrinsic nonlinear wave growth. The dependences shown in Figure 3 indicate that T_{rep} is approximately proportional to the injection timescale τ_D but inversely proportional to the nonlinear growth rate Γ_{NL} . We therefore fit T_{rep} using a linear combination of the two timescales τ_D and Γ_{NL}^{-1} :

$$\Omega_{e0}T_{\text{rep}} = C_{\tau}\Omega_{e0}\tau_D + C_{\Gamma}\Omega_{e0}\Gamma_{NL}^{-1} + C_0, \quad (14)$$

where C_{τ} and C_{Γ} are the fitted coefficients associated with τ_D and Γ_{NL}^{-1} , respectively, and C_0 is the intercept. The fit yields $C_{\tau} = 9.87 \times 10^{-1}$,

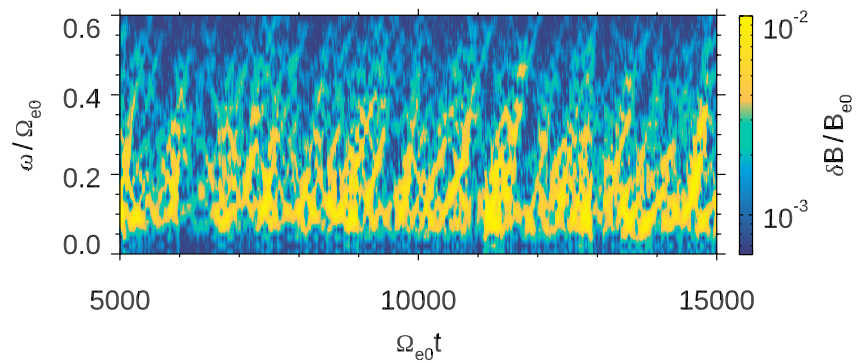


Figure 5. Simulated hiss-like chorus waves. Wave spectrum of $\delta B/B_{e0}$ at $\lambda = 5^\circ$ for the case with $n_e = 15 \text{ cm}^{-3}$ and $\Omega_{e0}\tau_D = 7,000$.

The role of field-line curvature can be investigated qualitatively through numerical simulations. In the dipole configuration used in this study, the effects of gradient and curvature cannot be separated. Kong, Gao, Ke, et al. (2023) instead employed a parabolic background magnetic field in a curved simulation domain, allowing curvature to vary while keeping the gradient fixed. They found that stronger field-line curvature facilitates chorus excitation and leads to larger wave amplitudes (Figure 7 in Kong, Gao, Ke, et al. [2023]), suggesting enhanced nonlinear growth in more stretched configurations.

In our simulations, decreasing η produces two competing effects: the increased magnetic-field gradient tends to reduce Γ_{NL} , whereas the enhanced field-line curvature tends to increase it. Their competition leads to the non-monotonic variation of T_{rep} with η . At relatively small η , the curvature effect appears to dominate, explaining why T_{rep} decreases despite the corresponding increase in $(\partial B_0/\partial h)^{1/2}$. These results highlight the importance of field-line curvature and motivate its incorporation into future nonlinear theories of chorus excitation.

Statistical observations indicate that T_{rep} is shorter on the nightside than on the dayside (Gao et al., 2022; Shue et al., 2015). This local-time asymmetry could be attributed to both plasma conditions and the magnetic-field configuration. For example, the higher normalized energetic-electron density n_{heq}/n_e on the nightside (Gao et al., 2014) can enhance the nonlinear growth rate. Meanwhile, the more stretched nightside magnetic-field configuration is associated with stronger field-line curvature, which may further promote chorus excitation. These combined effects tend to shorten the repetition period on the nightside.

From a physical perspective, nonlinear wave growth and energetic-electron injection play complementary roles in the release and replenishment of free energy. Nonlinear wave growth determines how rapidly free energy is converted into wave energy, such that a larger growth rate generally leads to more rapid chorus repetition under plasma conditions favorable for chorus excitation. Energetic-electron injection replenishes the depleted free energy and sustains chorus element repetition. In the absence of injection ($\tau_D \rightarrow \infty$), only a single element is typically excited under otherwise favorable conditions (Figure 2a of Lu et al. [2021]; Case II of X. Chen et al. [2026]). Therefore, both are required for the excitation of repetitive chorus elements. A special case occurs when injection is absent but nonlinear growth is sufficiently strong, producing a dense cluster of chorus elements (Case III of X. Chen et al. [2026]). This occurs because the excitation of one element does not exhaust all the free energy, allowing the remaining free energy to drive additional elements. Such clusters, however, differ markedly from repetitive chorus in observations: they typically contain only 2–3 elements, have very short repetition periods of $< 500 \Omega_{e0}^{-1}$ ($\sim 0.02 \text{ s}$ at $L \sim 6$), or even lack a well-defined repetition period, whereas repetitive chorus in magnetosphere commonly contains ~ 10 or more elements with repetition periods on the order of 0.1 s (Gao et al., 2022; Shue et al., 2015). In addition, the element amplitudes generally decrease as the free energy is depleted, unlike the comparable amplitudes often seen in observations.

When T_{rep} is sufficiently short, individual chorus elements become closely spaced in time, producing a hiss-like chorus spectrum. Previous studies suggested that rapid injection can lead to hiss-like chorus formation (Kong, Gao, Lu, & Wang, 2023). Here, we have demonstrated that T_{rep} is also regulated by the nonlinear growth rate. To examine this effect, we perform an additional simulation with $n_e = 15 \text{ cm}^{-3}$, which yields a relatively high

nonlinear growth rate, while adopting $\tau_D = 7,000 \Omega_{e0}^{-1}$, a relatively long injection timescale that reduces the influence of injection. Figure 5 shows closely spaced rising-tone elements, resembling a hiss-like chorus spectrum. Such high plasma densities can occur at $L \sim 6$, as illustrated by the event shown in Figure S2 of the Supporting Information S1. This simulation indicates enhanced nonlinear growth as a possible mechanism for hiss-like chorus formation, while event-specific simulations are left for future work.

The repetitive chorus emissions have also been reported in diverse planetary and lunar plasma environments, such as those near Mars (X. Ma et al., 2023), Jupiter (Q. Ma et al., 2024), Saturn (Hospodarsky et al., 2008), and the Moon (Sawaguchi et al., 2021). The mechanism identified in this study provides a theoretical framework for interpreting chorus element repetition beyond Earth's magnetosphere.

5. Summary

Using typical plasma parameters at $L \sim 6$, we perform GCPIC simulations to quantify the repetition period T_{rep} of chorus elements. A parameter survey involving nearly 50 simulation cases is conducted to examine the dependence of T_{rep} on plasma parameters and the background magnetic-field configuration. The results show that chorus element repetition is governed by two factors: external energetic-electron injection and intrinsic nonlinear wave growth. Specifically, T_{rep} increases with the electron injection timescale τ_D but decreases with the nonlinear growth rate Γ_{NL} . Regression analysis further indicates that nonlinear wave growth has a slightly greater influence on T_{rep} within the parameter ranges considered.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Availability Statement

The data used for plotting the figures are available at H. Chen and Wang (2026).

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